

SEINAJOKI PELMAA, Finland

WMO: 028330

Lat: 62.9381N

Lon: 22.4889E

Elev: 104

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-5.7	-17.7	2.1	-12.3	-10.1	3.2	-5.6	20.9	27.5	19.1	30.1	1.2	100	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	80.9	65.7	77.1	63.7	73.7	62.2	68.3	76.4	66.1	73.9	64.0	71.4	6.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.0	71.9	63.2	87.0	69.3	60.8	79.8	67.3	32.6	77.6	30.8	74.2	29.3	71.3	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.4	15.3	13.4	DB	-15.6	84.5	9.3	5.2	-22.3	88.2	-27.7	91.3	-32.9	94.2	-39.7	98.0	(4)
			WB	-14.0	70.6	10.8	4.0	-21.8	73.5	-28.1	75.8	-34.1	78.0	-42.0	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	19.0	21.4	27.7	38.3	49.5	56.4	62.0	58.9	50.9	39.8	32.9	25.2	(6)
(7)		DBStd	16.74	11.82	12.32	8.96	6.15	7.92	6.14	5.98	5.55	6.12	6.46	7.99	10.93	(7)
(8)		HDD50	4595	962	801	692	355	105	12	1	3	63	320	512	769	(8)
(9)		HDD65	9095	1427	1221	1157	802	484	269	131	204	422	782	962	1234	(9)
(10)		CDD50	1040	0	0	0	3	89	202	374	278	91	3	0	0	(10)
(11)		CDD65	66	0	0	0	0	3	10	39	14	0	0	0	0	(11)
(12)		CDH74	721	0	0	0	0	73	131	406	111	0	0	0	0	(12)
(13)		CDH80	137	0	0	0	0	5	26	88	18	0	0	0	0	(13)
(14)	Wind	WSAvg	5.6	5.6	6.2	6.2	5.6	5.8	5.0	4.3	4.8	5.0	6.0	6.4	6.4	(14)
(15)	Precipitation	PrecAvg	22.2	1.4	1.0	1.0	1.1	1.7	2.3	3.1	2.8	2.2	2.2	1.9	1.6	(15)
(16)		PrecMax	27.8	3.0	2.2	2.1	3.0	3.4	4.4	5.1	6.3	5.1	4.6	4.0	2.8	(16)
(17)		PrecMin	17.6	0.4	0.0	0.2	0.2	0.3	0.6	0.7	0.7	0.9	0.4	0.3	0.6	(17)
(18)		PrecStd	3.1	0.7	0.6	0.5	0.7	0.9	0.9	1.3	1.4	1.2	1.1	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	41.8	49.9	67.1	80.4	84.0	86.7	82.8	70.4	58.0	50.9	44.2	(19)
MCWB			38.0	38.2	41.6	51.9	60.8	65.5	69.3	68.7	60.7	53.8	48.7	43.1	(20)	
2%		DB	36.8	39.2	44.4	59.4	75.7	77.4	82.2	76.7	66.9	54.0	47.4	40.4	(21)	
		MCWB	35.2	36.4	39.0	48.8	58.7	62.9	67.7	64.3	58.8	51.1	45.6	38.4	(22)	
5%		DB	35.0	36.8	41.0	54.1	70.9	73.3	78.6	73.0	63.8	52.0	44.8	38.1	(23)	
		MCWB	34.0	35.3	36.9	45.3	57.0	60.3	65.5	62.3	56.8	49.4	43.4	36.6	(24)	
10%		DB	33.5	35.1	38.6	49.8	65.5	69.1	74.7	69.5	61.1	49.5	42.7	36.4	(25)	
		MCWB	32.7	34.0	35.2	42.9	54.1	59.1	64.2	60.5	56.0	47.2	41.9	35.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	38.9	42.6	53.3	63.0	67.9	72.7	70.7	62.3	55.0	48.9	43.2	(27)
MCDWB			39.4	41.4	47.1	64.4	74.7	77.6	82.9	79.4	68.2	57.3	50.6	43.9	(28)	
2%		WB	35.8	36.7	39.4	49.5	60.7	65.1	69.6	66.3	60.2	52.2	45.8	39.0	(29)	
		MCDWB	36.5	38.4	43.5	58.4	72.9	74.7	78.0	73.1	65.1	53.7	47.1	40.2	(30)	
5%		WB	34.1	35.5	37.5	45.9	58.0	62.3	67.4	64.0	58.2	50.0	43.6	36.8	(31)	
		MCDWB	34.9	36.8	40.5	52.5	69.7	70.8	75.1	70.4	62.2	51.6	44.6	37.7	(32)	
10%		WB	32.8	34.2	35.6	43.5	54.9	60.0	65.6	62.0	56.6	47.5	41.9	35.4	(33)	
		MCDWB	33.6	35.1	38.1	49.4	64.8	67.6	73.2	67.4	60.2	49.3	42.7	36.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.9	11.6	14.5	16.1	20.2	18.3	18.6	17.8	15.7	11.2	6.8	8.9	(35)
MCDBR			7.9	8.8	14.3	25.8	28.7	26.4	25.0	22.8	19.6	13.8	8.6	8.1	(36)	
MCWBR			7.4	7.4	9.8	16.0	14.2	13.6	12.1	11.8	11.9	10.9	7.9	7.6	(37)	
5% WB		MCDBR	7.6	8.1	13.0	23.2	27.0	22.1	21.8	19.2	16.8	12.5	8.4	8.2	(38)	
		MCWBR	7.3	7.0	9.6	15.1	14.0	11.9	11.5	10.8	11.1	10.7	8.0	7.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.295	0.320	0.359	0.355	0.353	0.375	0.361	0.336	0.318	0.258	0.352	(40)	
(41)		taud	2.058	2.185	2.276	2.333	2.424	2.450	2.411	2.444	2.504	2.381	2.137	2.570	(41)	
(42)		Ebn at Noon	125	202	247	261	273	276	265	259	244	196	116	70	(42)	
(43)		Edh at Noon	10	20	27	32	32	31	32	28	22	17	9	6	(43)	
(44)	All-Sky Solar Radiation	RadAvg	55	235	711	1177	1587	1725	1607	1231	722	283	71	26	(44)	
(45)		RadStd	7	50	73	128	169	150	155	128	80	56	13	3	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page